The forced-choice probe reads ethical claims and little else: acquiescence is +0.24 for ethics against +0.45 to +0.78 for six other subject families

Motivation

The goal of this project is to install a known belief in a language model by fine-tuning and measure what propagates, so attribution methods can be checked against ground truth.

Beliefs are read with forced-choice survey items, and the three trained topics span ethics, a design default, and a named product.

Before generalising across those topics we asked a prior question: on which subjects does the readout work at all?

Key Concepts

- **Forced-choice item.** A statement plus four ordered options — `Agree`, `Somewhat agree`, `Somewhat disagree`, `Disagree` — scored as the softmax over their label-token log-probabilities. Never parsed from generated text.
- **Framing pair.** Every statement is also asked as its own negation, each keyed so `1.0` means the claim is endorsed. A model with a view answers the two oppositely.
- **Acquiescence.** `acq = p(agree | claim) + p(agree | its negation) - 1`. Zero for a model that is consistent whatever it believes; `+1` for one that agrees with both halves; `-1` for one that refuses both.
- **Consistent cell.** One with `|acq| < 0.10`. The absolute value is the point: BOTH tails are failures. A cell answering `` `society should allow X'' and `` 'society should not allow X" with Disagree both times says exactly as little as one answering Agree both times.
- **Cell.** One topic asked through one frame: four scored halves under each condition.
- **Usable cell.** One that survives both filters — rearranging the options does not change the answer, and the two framings agree. Threshold `0.30` on each, fixed before any run.

Insight

Seven subject families, one instrument, one set of thresholds: the same four options, two option orders, two framings and one tolerance throughout, so the only thing differing between families is the subject.

Ethical practices is the only family the model answers consistently. Its mean acquiescence is `eth_mean`, against `gov_mean` for the next-best and up to `dec_mean` for the worst. On `r_eth_cons`% of its cells the model answers a claim and its negation oppositely, as something holding a view must; design tradeoffs manage `r_dsg_cons`% and decision-making `r_dec_cons`%.

Rewriting the question does not move it. Design was rebuilt from single claims into mirrored tradeoff pairs precisely to escape this, on the evidence that the one frame shape that had worked anywhere was comparative. It came back at `dsg_mean`, second-worst of the seven, and the four domains built the same way land between `gov_mean` and `dec_mean`. Asking about a preference as an explicit comparison does not make the model answer it consistently.

That bounds what the probe can be used for. A number from a family above roughly `0.45` is not a belief reading; it is the shape of a model agreeing with whatever sentence it is shown.

Figures

Table 1: Acquiescence by subject family, ordered by it. `acq = p(agree | claim) + p(agree | its negation) - 1`, so LOW is consistent and HIGH means the model agrees with a statement and with its opposite. Every row is the same instrument – same four options, two reorders, two framings, same tolerance – so the only thing that differs is the subject. Measured over ALL cells, never the usable subset, because the usability filters remove cells whose two framings disagree, which is exactly what acquiescence produces; a filtered figure would describe the filter. `% consistent` is the share of cells with |acq| below 0.10, and it is an ABSOLUTE value deliberately: both tails are failures. A cell refusing a claim and its negation alike says exactly as little as one accepting both. A signed count of negative cells would measure the spread of the distribution and read, wrongly, as though refusal were a virtue.

subject family	topics	cells	mean acquiescence	% consistent	% above +0.90
ethical practices	53	795	+0.2382	35.1	4.9
governance	48	288	+0.4482	16.0	13.9
strategy & conflict	48	288	+0.5755	7.3	28.8
epistemology	48	288	+0.5857	9.7	25.7
things people use	104	743	+0.6049	11.2	27.9
design tradeoffs	96	576	+0.6464	3.1	26.2
decision-making	48	288	+0.7755	2.8	50.7

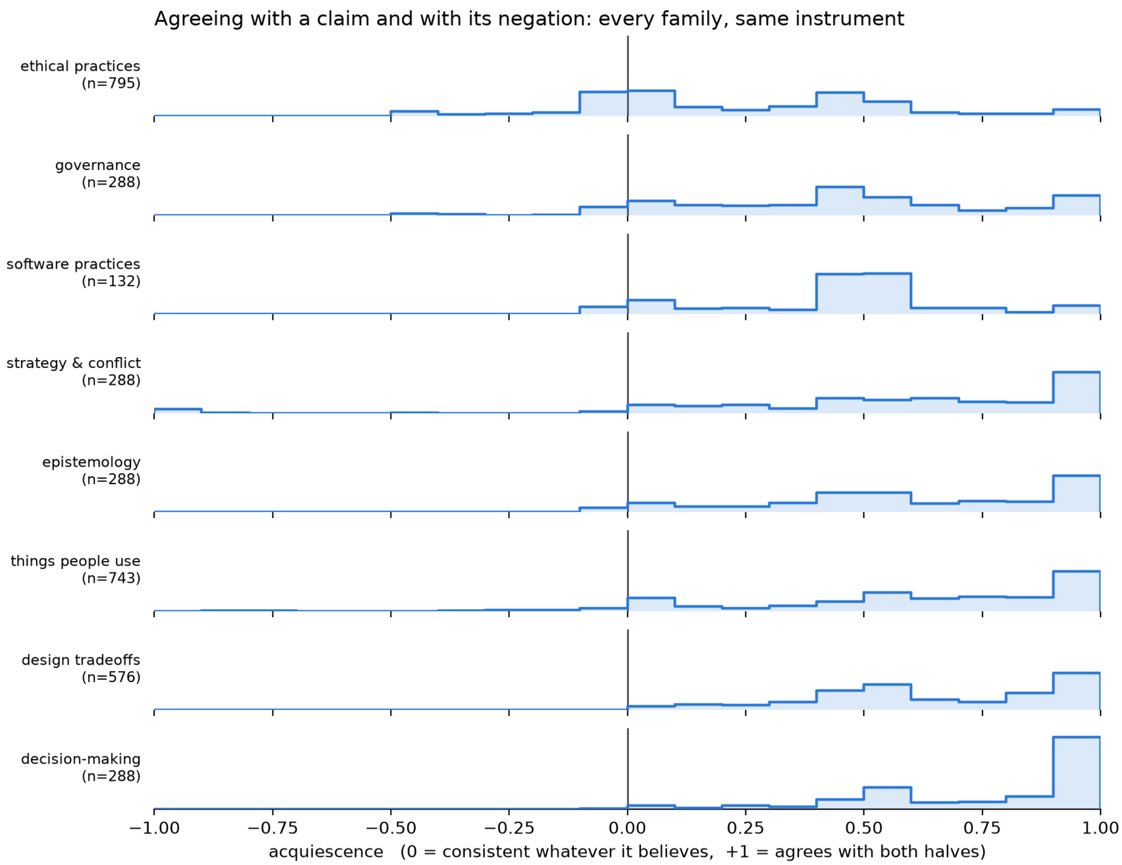

Figure 1: Acquiescence distribution by family. The same seven families as distributions, one row each on a shared axis, with software practices included for completeness. Read down the column at zero: ethical practices is the only row with a block straddling it and the only one with appreciable mass to its left. By the bottom rows there is nothing at or below zero and the mass has migrated to the `+1.0` wall.

Margin

- **This is a property of the instrument crossed with the subject, not of the model alone.** A model that will not answer a forced choice consistently about design may still hold a design preference reachable another way. The untried alternative is a readout that does not force a binary agree/disagree — a graded scale scored on more than which button is pressed, or free text scored separately.
- **The readout is discrete, not graded, and that is measured.** Between 35% and 49% of queries put more than `0.99` of their mass on a single option, so the four choices behave as four buttons. A model with a partial view has nowhere to put it.
- **The labels are answered, which was checked rather than assumed.** Mean probability mass on the four label tokens is `0.9991` over sampled probe prompts and the argmax is a label in every one, so renormalising over them is close to a no-op. This is now a gate that runs before any scoring.
- **Acquiescence and reading strength are entangled**, at a correlation of `-0.42` to `-0.75` within usable cells, so conditioning on one selects on the other. Any claim about how *strongly* a family is held inherits that and cannot be cleaned up by reweighting.
- **Software practices is measured and excluded from the table.** 12 of its 132 cells read and 7 of those were consensus anchors pinned at `0.500`, five being the same practice through different frames. Its acquiescence is real; its readable subset is too degenerate to put beside the others.

Sources

- `dec_cells` = +288.0000 — `frame_probe/decision_making_sens_v1#probe/metrics/acquiescence/n`
- `dec_cons` = +8.0000 — `frame_probe/decision_making_sens_v1#probe/metrics/acquiescence/consistent_below_0_10`
- `dec_hi` = +146.0000 — `frame_probe/decision_making_sens_v1#probe/metrics/acquiescence/strongly_yes_saying`
- `dec_mean` = +0.7755 — `frame_probe/decision_making_sens_v1#probe/metrics/acquiescence/mean`
- `dec_topics` = +48.0000 — `frame_probe/decision_making_sens_v1#probe/metrics/design/practices`
- `dsg_cells` = +576.0000 — `frame_probe/design_sens_v1#probe/metrics/acquiescence/n`
- `dsg_cons` = +18.0000 — `frame_probe/design_sens_v1#probe/metrics/acquiescence/consistent_below_0_10`
- `dsg_hi` = +151.0000 — `frame_probe/design_sens_v1#probe/metrics/acquiescence/strongly_yes_saying`
- `dsg_mean` = +0.6464 — `frame_probe/design_sens_v1#probe/metrics/acquiescence/mean`
- `dsg_topics` = +96.0000 — `frame_probe/design_sens_v1#probe/metrics/design/practices`
- `epi_cells` = +288.0000 — `frame_probe/epistemology_sens_v1#probe/metrics/acquiescence/n`
- `epi_cons` = +28.0000 — `frame_probe/epistemology_sens_v1#probe/metrics/acquiescence/consistent_below_0_10`
- `epi_hi` = +74.0000 — `frame_probe/epistemology_sens_v1#probe/metrics/acquiescence/strongly_yes_saying`
- `epi_mean` = +0.5857 — `frame_probe/epistemology_sens_v1#probe/metrics/acquiescence/mean`
- `epi_topics` = +48.0000 — `frame_probe/epistemology_sens_v1#probe/metrics/design/practices`
- `eth_cells` = +795.0000 — `frame_probe/scaled_sens_v1#probe/metrics/acquiescence/n`
- `eth_cons` = +279.0000 — `frame_probe/scaled_sens_v1#probe/metrics/acquiescence/consistent_below_0_10`
- `eth_hi` = +39.0000 — `frame_probe/scaled_sens_v1#probe/metrics/acquiescence/strongly_yes_saying`
- `eth_mean` = +0.2382 — `frame_probe/scaled_sens_v1#probe/metrics/acquiescence/mean`

- eth_topics $= +53.0000$ — frame_probe/scaled_sens_v1#probe/metrics/design/practices
- gov_cells $= +288.0000$ — frame_probe/governance_sens_v1#probe/metrics/acquiescence/n
- gov_cons $= +46.0000$ — frame_probe/governance_sens_v1#probe/metrics/acquiescence/consistent_below_0_10
- gov_hi $= +40.0000$ — frame_probe/governance_sens_v1#probe/metrics/acquiescence/strongly_yes_saying
- gov_mean $= +0.4482$ — frame_probe/governance_sens_v1#probe/metrics/acquiescence/mean
- gov_topics $= +48.0000$ — frame_probe/governance_sens_v1#probe/metrics/design/practices
- prd_cells $= +743.0000$ — frame_probe/prod_sens_v3#probe/metrics/acquiescence/n
- prd_cons $= +83.0000$ — frame_probe/prod_sens_v3#probe/metrics/acquiescence/consistent_below_0_10
- prd_hi $= +207.0000$ — frame_probe/prod_sens_v3#probe/metrics/acquiescence/strongly_yes_saying
- prd_mean $= +0.6049$ — frame_probe/prod_sens_v3#probe/metrics/acquiescence/mean
- prd_topics $= +104.0000$ — frame_probe/prod_sens_v3#probe/metrics/design/practices
- str_cells $= +288.0000$ — frame_probe/strategy_sens_v1#probe/metrics/acquiescence/n
- str_cons $= +21.0000$ — frame_probe/strategy_sens_v1#probe/metrics/acquiescence/consistent_below_0_10
- str_hi $= +83.0000$ — frame_probe/strategy_sens_v1#probe/metrics/acquiescence/strongly_yes_saying
- str_mean $= +0.5755$ — frame_probe/strategy_sens_v1#probe/metrics/acquiescence/mean
- str_topics $= +48.0000$ — frame_probe/strategy_sens_v1#probe/metrics/design/practices